Rakesh varma pericherla

TECHNICAL LEAD

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SKILLS

- Problemsolving abilities
- C++
- QT
- QML
- Android
- Object Oriented Programming
- Sqlite
- MySql
- Android
- iOS
- Java

LANGUAGES

English

Hindi

Telugu

kannada

SUMMARY

With 9.5 years of extensive experience in software development, I specialize in C++ and Qt/QML technologies. My role spans multiple facets, including designing user interfaces, integrating systems, and enhancing application performance. I have a proven track record in developing real-time applications for both embedded systems and mobile platforms. My commitment to quality and innovation ensures successful project outcomes.

EXPERIENCE

Technical Lead — Mar 2022 - Nov 2024
Harman Connected services

Senior Software Engineer — Jan 2021 - Feb 2022
Embitel Technologies Pvt Ltd

Lead Engineer — Oct 2019 - Jan 2021
HCL Technologies

Senior Software Engineer — Jun 2018 - Aug 2019
Cadmaxx solutions Pvt Ltd (Client: Aptiv Technical center)

Senior Software Engineer — Nov 2014 - Nov 2017
Mistral solutions Pvt Ltd

Intern — Feb 2014 - Aug 2014
CDAC-ACTS

EDUCATION

Bachelor of Engineering and Technology — Aug 2008 - Jun 2012
SMIST

MPC — Apr 2006 - Apr 2008
Narayana Junior College, Vishakapatnam

SSC — Apr 2005 - Apr 2006
Bhashyam Public school, Vishakapatnam.

PROJECTS

ArrayLink — Mar 2022 - Nov 2024
Developer

Environment : C++, Qt/QML

Database : SQLITE

Operating system : Android, iOS

JBL Array Link is a mobile companion app that works in conjunction with JBL's latest LAC-III software to assist technicians deploying JBL VTX Series line array systems. Array Link uses a QR code system to transfer all array mechanical information from the main LAC-III software application to a mobile phone – this transfer is done directly and in real-time without the need for internet connection. All relevant rigging information and options are presented in an easy-to-understand layout.

Roles/Responsibilities:

- C++ based interface implementation with GUI designed using QML
- Enhanced the application GUI with latest Qt features

JBL Pro Connect — Mar 2022 - Nov 2024

Developer

Environment : C++, Qt/QML

Database : SQLITE

Operating system : Android, iOS

Description: Control JBL Professional PRX ONE, EON700 or EON ONE MK2 portable PA speakers directly from your Android device with the free JBL Pro Connect app. Using Bluetooth® Low Energy (BLE) technology, JBL Pro Connect gives you complete hands-on control over all of your PA system's built-in mixer and DSP functions. Build your ideal PA system by syncing up to 10 speakers. JBL Pro Connect gives you all of the tools to sound your best, right at your fingertips. Dial in your signature sound from anywhere in the room: Set volume levels, adjust EQ settings, activate Lexicon reverb and effects and dbx Digital DriveRack signal processing, customize ducking, save and recall presets and much more. Whether you're a seasoned musician, DJ or presenter or just starting out on your live sound journey, JBL Pro Connect makes it easy to tailor your tones for any venue or application. FULL CONTROL, IN THE PALM OF YOUR HAND · Manage JBL PRX ONE, EON700 or EON ONE MK2 portable PA speakers from your android phone or tablet · Dive deep into system mixer and DSP functions to optimize your stage sound for any scenario · Access app-exclusive features and speaker settings including snapshots, multi-speaker grouping, tap tempo and more ELEVATE YOUR STAGE SOUND WITH PRO EFFECTS · Activate ducking to make sure your voice stands out over background music · Adjust Lexicon reverb, chorus and delay effects for professional polish · Shape sound with surgical precision with parametric EQ SOUND AMAZING IN AN INSTANT · Choose from eight optimized sound presets, or customize your own · Simply and quickly adjust master volume and source levels · Trigger dbx Automatic Feedback Suppression (AFS) to eliminate feedback · Optimize and protect speakers with dbx Digital DriveRack signal pr

Roles/Responsibilities:

- C++ based interface implementation with GUI designed using QML
- Enhanced the application GUI with latest Qt features.

Edison PHY Development(Etn Eval) — Jan 2021 - Feb 2022

Developer

Environment : PyQt5, Qt/QML

Database :

Operating system : Linux(ubuntu 21.04)

Description: The project involves the software development of beaglebone board, which is used to develop, test and debug Edison chips used for High speed ethernets. The project contains the driver, middleware and application layers. Roles/Responsibilities:

- Python based interface implementation with GUI designed using QML
- Enhanced the application GUI with latest Qt features.

VP2, CNHI (HMI Development) **Jun 2018 - Aug 2019**
Developer

Environment : C++, Qt/QML

Database :

Operating system : Linux, QNX

Description: This project is Radio Infotainment on Daily hardware with QNX 6.5.0 RTOS. HMI is developed on Qt 5.3 which is built for QNX. IPC using QNX Message Queues to communicate with middleware / functional components. HMI generates an event to the model when user performs an action. This project contains the QT QML based UI to show the different entertainment sources like USB_MSD, USB_MTP, Disk, IPOD, Projection, Bluetooth, Native maps, user can select the source from source selection and play the songs.

Roles/Responsibilities:

- Analysis of requirements from IBM Rational Doors.
- Coding, and Integration of the requirement on infotainment boards.
- To understand the lower block API's that is consumed by HMI.
- Worked on Qt/QML to develop the device screens.
- Did feature addition and bug-fixes.
- Testing the developed HMI on hardware provided by client

IWTP (Integrated Wheeled Test Platform)
Developer

Environment : C++ , Python

Database : MySQL

Operating system : Linux, ROS

Web Technologies : JavaScript, HTML, CSS Description: Integrated Wheeled Test Platform (IWTP) is a battery operated unmanned ground vehicle for patrolling and first response. It is power packed to operate for 8 hours and ruggedized to operate for all-weather, paved on-road and unpaved off-the-road conditions. IWTP is designed to operate in manual and drive-by-wire modes. It is mounted with sensors and embedded electronics, which provides the necessary data for the drive-by-wire system to operate in Tele-operation and autonomous mode of navigation. In the Tele-operation mode, the user operates the IWTP using the driving controls from the Master Control Station (MCS). In the autonomous mode of navigation, IWTP navigates to a designated point or patrol in a cyclic path. IWTP is operated either through MCS or on-board display present on the vehicle and is designed to support following features in autonomous mode of operation Path Planning, Environment mapping,Localization,Obstacle avoidance. 5 | P a g e DC MIST Project :Smart Energy Meter (AWS IOT) Project : Industrial Power Quality Analyzer User can access the Integrated Wheeled Test

Platform(IWTP) using web browser from various operating systems like Windows, Linux, Android and MAC OSX. Roles/Responsibilities:

- Involved in requirements capturing
- Database design for IWTP
- Design and Development of Python based web interface for IWTP.
- Developed test case document to test entire web interface.

Industrial Power Quality Analyzer
Developer

Environment : C++, Qt, QWidgets

Database : MySQL

Operating system : Linux

Description: This is a Class A power quality analyzer according to IEC 61000-4-30 Ed.3. In addition to perform basic measurements such as voltage, current and power on a three-phase line, it will also measure harmonics and detect power quality events. It will generate power quality reports according to EN50160. The product will have TFT LCD with touch-panel and 5 HW buttons its runs the QT based embedded GUI application, Web based User interface, IEC 61850 (over TCP/IP), Modbus RTU (over RS-485), Modbus TCP (over Ethernet), VPN, Embedded database, Digital I/O, Analog and relay outputs, Time synchronization over SNTP and IRIG-B.

Roles / Responsibilities:

- Design and development of QT embedded application.
- IO_INIT Application.
- Doxygen documentation of the developed code and logics.

MindGraph
Developer

Environment : Android (Java)

Database : Mysql

Operating system : Linux

Description: The MindGraph is a Medical Device with Android KITKAT V4.4, Where ECG/EEG data is collected from the patient using knobs. Patient Data is collected using 3 ADS1299 chips. The Medical Device is Interfaced with 3 ADC (ADS1299) chips, NHD Display, FocalTech Touch. ADC data is collected from the patient from 8 channels of each chip using ads1299 driver and the collected data is given to the Application.There is a custom Android Application that takes the data from driver and display it Accordingly.

Roles / Responsibilities:

- Involved in design, development and testing of the Ndk based Android application.
- Developed test case document to test entire architecture.
- Develop the Software Black box test cases for Functional Specification using Robotium.
- Generate the functional test report and coverage report.

OTHER DETAILS

- Date Of Birth: 14 July 1991
- Marital Status: Single
- Current Location: Bangalore

- Home Town: Payakaraopeta, Andhra Pradesh

Place : Bangalore

Date: (Rakesh Varma P)